

Sharp Conditioning Laws for Fabius and Lacunary Random Series

A global simplex approximation, exact total-variation thresholds,
and phase-dependent boundary layers

Research manuscript prepared for Vladimir Reshetnikov
AI-assisted derivations, with reproducible numerical checks

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Abstract

Let $S = \sum_{j \geq 1} w_j U_j$, where the U_j are independent uniforms on $[0, 1]$ and $0 < w_{j+1} \leq q w_j$ for a fixed $q < 1$. We investigate the entire configuration conditioned on $S \leq x$, as $x \downarrow 0$. Write t for the exponential-tilting parameter whose tilted mean is x , and $n = \#\{j : t w_j \geq 1\}$. We prove a quantitative approximation, in total variation on the full countable product space, by independent Dirichlet proportions, the tilted tail, and an exponential slack, followed by a common radial rescaling. Its error is $O_q((1 + \log \log n)/n)$.

For an arbitrary, possibly infinite coordinate set containing a proportion α of the active coordinates, the distance between its conditioned and tilted laws tends to $d_{TV}(\mathcal{N}(0, 1 - \alpha), \mathcal{N}(0, 1))$. Thus negligible tilted variance is exactly the criterion for asymptotic independence from the constraint. For $k = o(n)$ growing bulk coordinates, the distance is $\varphi(1)k/n + o(k/n)$; an explicit L^1 density expansion also gives fixed-coordinate mean and covariance corrections. We obtain a Brownian-bridge limit, exponential endpoint slack, Gumbel extremes, and an explicitly phase-dependent microscopic boundary law for geometric weights. The dyadic case is the Fabius distribution.

The finite-i.i.d. total-variation phenomenon belongs to classical work of Diaconis–Freedman. The contribution developed here is its quantitative infinite-lacunary extension, especially the full-configuration approximation and the simultaneous control of bulk, slack, and boundary. These are manuscript proofs, not Lean-certified results; priority beyond the inspected sources is not asserted.

Repository anchor. VladimirReshetnikov/ProveIt, Analysis/FabiusFunction.
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1 Research target, provenance, and scope

The probabilistic construction of the Fabius function asks about the small values of

$$S_2 = \sum_{j \geq 1} 2^{-j} U_j, \quad F_2(x) = \mathbb{P}(S_2 \leq x). \quad (1.1)$$

This representation is classical; Arias de Reyna's Theorem 3 gives it in the normalization $F_2(x) = \text{up}(x - 1)$ for $0 \leq x \leq 1$ [1]. The question studied here is not only *how unlikely* $S_2 \leq x$ is, but *what the infinitely many summands look like when that event occurs*.

The inspected ProveIt asymptotic audit reports an all-orders small-argument expansion, with an exact Lambert phase and periodic correction terms [4]. We do not re-prove that expansion or claim to have rebuilt its Lean files. Instead we address three related questions. How many summands may be inspected before the conditioning becomes statistically detectable? Can one approximate the *entire* conditioned configuration without making an incorrect independence assertion? Where do the geometric oscillations survive in a probabilistic limit?

Theorems below answer these questions for every uniformly lacunary sequence, not just powers of two. The main mechanism is simple to state but requires uniform error control: sufficiently early tilted summands are nearly exponentials; their proportions form a simplex; and the remaining coordinates have only $O(\log \log n)$ total tilted variance after a suitable split.

1.1 What is classical, and what is being extended

Diaconis and Freedman studied conditional laws of finite i.i.d. exponential-family samples and obtained sharp variation asymptotics for both $k/n \rightarrow 0$ and proportional blocks [2]. In particular, a linear k/n threshold and a Gaussian proportional-block profile must not be advertised as new finite-sample principles. We derive the needed simplex formulas explicitly, with our convention $d_{\text{TV}} = \frac{1}{2} L^1$, and extend them to an infinite, non-identically distributed, increasingly tilted array.

Cator and Ferrari's more recent Gibbs-conditioning theorem concerns independent nonnegative integer-valued log-concave laws, a fixed tilt, and weak convergence under an anti-condensation hypothesis [3]. Our variables are continuous, the tilt diverges, the event is a shrinking small ball, and the main approximation is on a full countable product in total variation. This comparison identifies different hypotheses and conclusions; it is not a claim that existing general conditioning theory cannot also be adapted.

The repository review was targeted: its root map, Fabius asymptotic audit, and narrow code searches were inspected. It was not an exhaustive reading of every repository manuscript, nor an exhaustive priority search. Accordingly, we describe the results as a proved manuscript-level extension and a candidate contribution to this research program, not as a certified historical breakthrough or a resolution of a named, published open conjecture.

1.2 Conventions and proof organization

For probability measures on the same measurable space,

$$d_{\text{TV}}(P, Q) = \sup_A |P(A) - Q(A)| = \frac{1}{2} \int \left| \frac{dP}{dQ} - 1 \right| dQ \quad (P \ll Q), \quad D_{\text{KL}}(P \| Q) = \int \log \frac{dP}{dQ} dP.$$

Products over countably many coordinates carry the product Borel σ -field. All limits are as $x \downarrow 0$, equivalently $t \rightarrow \infty$. Constants with a subscript q may depend on the fixed lacunarity bound but not on the particular

weights, the observed coordinate set, or t , once n is sufficiently large. The normal density and distribution function are φ and Φ .

Sections 2–4 prove the full-configuration theorem. Sections 5–6 establish sharp marginal and information-theoretic consequences. Sections 7–9 handle processes and extremes. The numerical checks and open research agenda are separated from the proofs.

2 The tilted array and its effective dimension

Assume throughout that

$$w_j > 0, \quad w_{j+1} \leq q w_j \quad (j \geq 1), \quad 0 < q < 1. \quad (\text{L})$$

No lower bound on w_{j+1}/w_j is imposed. Thus very large gaps are allowed. Let P be the original product law and put

$$S = \sum_{j \geq 1} w_j U_j, \quad F(x) = P(S \leq x), \quad L(t) = \mathbb{E}_P e^{-tS} = \prod_{j \geq 1} \frac{1 - e^{-t w_j}}{t w_j}.$$

The series is bounded by $\sum_j w_j < \infty$, so these definitions pose no integrability difficulty. For $0 < x < \mathbb{E}_P S$ there is a unique $t > 0$ satisfying

$$x = \mathbb{E}_{Q_t} S, \quad \frac{dQ_t}{dP} = \frac{e^{-tS}}{L(t)}. \quad (2.1)$$

Indeed, the tilted mean is continuous, its derivative is $-\text{Var}_{Q_t}(S) < 0$, its value at zero is $\mathbb{E}_P S$, and it tends to zero. The last assertion follows by dominated convergence coordinate by coordinate, using summability of the weights. Also $F(x) > 0$: choose a deterministic tail bound below $x/2$, and then restrict finitely many uniforms to sufficiently small positive intervals.

Define scaled coordinates, their caps, and the active dimension by

$$Y_j = t w_j U_j, \quad a_j = t w_j, \quad T = \sum_{j \geq 1} Y_j, \quad \mu = t x = \mathbb{E}_{Q_t} T, \quad n = \#\{j : a_j \geq 1\}. \quad (2.2)$$

The number n is finite for fixed t and tends to infinity with t . Under Q_t the coordinates are independent, with densities

$$p_a(y) = \frac{e^{-y}}{1 - e^{-a}} \mathbf{1}_{(0,a)}(y), \quad m(a) = 1 - \frac{a}{e^a - 1}, \quad v(a) = 1 - \frac{a^2 e^a}{(e^a - 1)^2}. \quad (2.3)$$

Here m and v are the mean and variance. Their values near zero are understood by continuous extension; in particular $m(a) = a/2 + O(a^2)$ and $v(a) = a^2/12 + O(a^4)$.

Lemma 2.1 (Active-dimension estimates). *Uniformly under (L),*

$$\mu = n + O_q(1), \quad V := \text{Var}_{Q_t}(T) = n + O_q(1). \quad (2.4)$$

Moreover, uniformly for every coordinate set $I \subseteq \mathbb{N}$,

$$\sum_{j \in I} v(a_j) = |I \cap [1, n]| + O_q(1). \quad (2.5)$$

Proof. For $r \geq 0$, $a_{n-r} \geq q^{-r}$, whereas for $r \geq 1$, $a_{n+r} < q^{r-1}$. The active mean deficits $1 - m(a)$ and variance deficits $1 - v(a)$ are nonnegative and decay exponentially in a up to polynomial factors. Their sums are bounded by convergent series evaluated at q^{-r} . For the inactive tail, $0 \leq m(a) \leq a$ and $0 \leq v(a) \leq a^2$, so its mean and variance sums are bounded by geometric series. This proves (2.4). The same argument bounds the sum of absolute discrepancies $\sum_j |v(a_j) - \mathbf{1}_{j \leq n}|$ by a constant; restricting the sum to any I gives (2.5). $\square$

Let $P_x = P(\cdot \mid S \leq x)$ and define its nonnegative slack by

$$H = \mu - T = t(x - S). \quad (2.6)$$

The exact change of measure is

$$\frac{dP_x}{dQ_t} = \frac{e^{T-\mu} \mathbf{1}_{\{T \leq \mu\}}}{D_t}, \quad D_t = \mathbb{E}_{Q_t} [e^{T-\mu} \mathbf{1}_{\{T \leq \mu\}}], \quad F(x) = L(t) e^\mu D_t. \quad (2.7)$$

The weight in the numerator is at most one. This elementary fact will allow an exponentially accurate product approximation to survive conditioning on an event whose tilted normalizing mass is only of order $n^{-1/2}$.

3 Separating an exponential bulk from the tail

For all sufficiently large n , set

$$\ell = \left\lceil \frac{\log(6 \log n)}{\log(1/q)} \right\rceil, \quad N = n - \ell, \quad \rho_n = \frac{\ell + 1}{n}. \quad (3.1)$$

Then $\ell = O_q(1 + \log \log n)$, $N \geq n/2$ eventually, and $a_N \geq 6 \log n$. Write

$$Z = (Y_j)_{j > N}, \quad R = \sum_{j > N} Y_j, \quad r = \mathbb{E}_{Q_t} R, \quad W = R - r, \quad \varepsilon = \sum_{j \leq N} (1 - m(a_j)). \quad (3.2)$$

In particular,

$$\mu = N - \varepsilon + r. \quad (3.3)$$

Lemma 3.1 (Tail moments and removable caps). *The above quantities satisfy*

$$r + \mathbb{E}W^2 = O_q(\ell + 1), \quad \mathbb{E}W^4 = O_q((\ell + 1)^2), \quad (3.4)$$

$$\varepsilon = O_q((\log n)n^{-6}), \quad \delta := \sum_{j \leq N} e^{-a_j} = O_q(n^{-6}). \quad (3.5)$$

For each fixed even integer p , $\mathbb{E}|W|^p = O_{q,p}((\ell + 1)^{p/2})$.

Proof. There are ℓ remaining active coordinates. For $a \geq 1$, the density p_a is bounded by $(1 - e^{-1})^{-1} e^{-y}$, and hence every fixed raw or centered moment is uniformly bounded. For $j > n$, the cap is bounded by a geometric sequence, so the sums of all fixed positive powers of the caps are bounded. This gives the mean and variance estimates.

For independent centered variables X_j , direct expansion gives

$$\mathbb{E}\left(\sum_j X_j\right)^4 = \sum_j \mathbb{E}X_j^4 + 6 \sum_{i < j} \mathbb{E}X_i^2 \mathbb{E}X_j^2.$$

The first sum is $O_q(\ell + 1)$ and the second is $O_q((\ell + 1)^2)$. The same expansion for any fixed even p has no term containing a singleton index, so at most $p/2$ distinct indices occur. The moment and geometric-tail bounds give the stated general estimate. Finite partial sums suffice initially; the bounded tail permits passage to the limit.

For $A = a_N$ and $c = q^{-1} - 1 > 0$, we have $a_{N-s} \geq Aq^{-s} \geq A(1 + cs)$. Consequently

$$\sum_{j \leq N} e^{-a_j} \leq \frac{e^{-A}}{1 - e^{-cA}} = O_q(n^{-6}).$$

The same exponentially decreasing comparison, with an extra factor a_j , bounds $\sum_{j \leq N} a_j / (e^{a_j} - 1)$ by $O_q(Ae^{-A})$. Since $A \geq 6 \log n$ and Ae^{-A} is decreasing for $A \geq 1$, this is $O_q((\log n)n^{-6})$. $\square$

Replace the first N coordinates under Q_t by independent untruncated $\text{Exp}(1)$ variables, leaving the whole tail law unchanged. Denote the resulting product measure by $\tilde{Q}_t$. With

$$B = \{Y_j < a_j \text{ for all } j \leq N\},$$

we have the exact relation

$$Q_t = \tilde{Q}_t(\cdot | B), \quad \tilde{Q}_t(B^c) \leq \delta. \quad (3.6)$$

Define $\tilde{P}_x$ by weighting $\tilde{Q}_t$ with the same function $e^{T-\mu} \mathbf{1}_{T \leq \mu}$, and let its normalizer be $\tilde{D}_t$. Then

$$P_x = \tilde{P}_x(\cdot | B), \quad d_{\text{TV}}(P_x, \tilde{P}_x) = \tilde{P}_x(B^c) \leq \delta / \tilde{D}_t. \quad (3.7)$$

Here and below the corresponding slack coordinate can be adjoined without increasing the distance, because it is a measurable function of the configuration on either weighted law's support.

4 A full-configuration simplex approximation

For integer $M \geq 1$, let

$$g_M(s) = \frac{e^{-s} s^{M-1}}{\Gamma(M)} \mathbf{1}_{\{s>0\}}$$

be the gamma density, and put

$$c_M = g_M(M) = g_{M+1}(M) = \frac{e^{-M} M^M}{M!}.$$

Stirling's formula gives $c_M = (2\pi M)^{-1/2} (1 + O(M^{-1}))$.

Lemma 4.1 (Flatness on the scale of the tail). *For $M \geq 2$ and real d ,*

$$0 \leq \frac{g_M(M+d)}{c_M} \leq e, \quad \left| \frac{g_M(M+d)}{c_M} - 1 \right| \leq C \frac{|d| + d^2}{M}. \quad (4.1)$$

Proof. For $d > -M$, the logarithm of the ratio is $(M-1) \log(1 + d/M) - d$. Its maximum occurs at $d = -1$ and is at most one. For $|d| \leq \sqrt{M}$ and sufficiently large M , Taylor's formula bounds the absolute logarithm by $C(|d| + d^2)/M$; the logarithm is bounded on this interval, so the same estimate holds for the difference of the ratio from one. For $|d| > \sqrt{M}$, the right side is bounded below by an absolute positive constant and the already proved upper bound suffices. The finitely many small values of M are covered by increasing C . When $d \leq -M$, the ratio is zero and the inequality is immediate. $\square$

Under $\tilde{Q}_t$, the bulk sum $G = \sum_{j \leq N} Y_j$ has density g_N . Its proportions $(Y_1/G, \dots, Y_N/G)$ have distribution $\text{Dir}(1, \dots, 1)$ and are independent of G and the tail. This follows directly from the change of variables $y_j = g d_j$, whose Jacobian on the simplex is g^{N-1} . The joint density factors into a function of g and a constant function of the proportions.

Under $\tilde{P}_x$ the proportions remain independent. Changing variables from G to $H = \mu - R - G$ shows that the joint law of (Z, H) has density

$$\frac{g_N(\mu - R - h)}{\tilde{D}_t} \quad \text{with respect to} \quad Q_t^{>N}(dz) e^{-h} \mathbf{1}_{h \geq 0} dh. \quad (4.2)$$

This is an exact identity, not a limiting heuristic.

Theorem 4.2 (Global conditioned-configuration approximation). *Independently sample*

$$Z \sim Q_t^{>N}, \quad E \sim \text{Exp}(1), \quad D = (D_1, \dots, D_N) \sim \text{Dir}(1, \dots, 1).$$

Let $R = \sum_{j>N} Z_j$, set $A = (\mu - R - E)_+$, and define

$$Y_j^* = AD_j \quad (j \leq N), \quad Y_j^* = Z_j \quad (j > N). \quad (4.3)$$

Then, on $[0, \infty)^{\mathbb{N}} \times [0, \infty)$,

$$d_{\text{TV}}\left(\mathcal{L}_{P_x}((Y_j)_{j \geq 1}, H), \mathcal{L}((Y_j^*)_{j \geq 1}, E)\right) \leq C_q \rho_n. \quad (4.4)$$

Furthermore,

$$D_t = \frac{1}{\sqrt{2\pi n}}(1 + O_q(\rho_n)), \quad d_{\text{TV}}(P_x, \tilde{P}_x) = O_q(n^{-11/2}). \quad (4.5)$$

Proof. In (4.2), use (3.3) to write

$$\mu - R - h = N + d, \quad d = -W - h - \varepsilon.$$

Under the reference product law of (Z, E) , Lemma 3.1 gives $\mathbb{E}(|d| + d^2) = O_q(\ell + 1)$. Lemma 4.1 therefore yields

$$\mathbb{E} \left| \frac{g_N(\mu - R - E)}{c_N} - 1 \right| \leq C_q \frac{\ell + 1}{N}.$$

Taking expectations and then normalizing gives

$$\tilde{D}_t = c_N(1 + O_q((\ell + 1)/N)), \quad d_{\text{TV}}(\mathcal{L}_{\tilde{P}_x}(Z, H), Q_t^{>N} \otimes \text{Exp}(1)) = O_q((\ell + 1)/N). \quad (4.6)$$

The independent proportions can be appended to both laws. Applying the measurable reconstruction map (4.3) cannot increase total variation. The exact law in (4.2) puts zero mass where $\mu - R - h \leq 0$, so the positive-part convention agrees with exact reconstruction on its support.

Now (3.7) and Lemma 3.1 give $d_{\text{TV}}(P_x, \tilde{P}_x) = O_q(n^{-6}\sqrt{N}) = O_q(n^{-11/2})$. This proves (4.4). The exact identity

$$D_t = \tilde{D}_t \frac{\tilde{P}_x(B)}{\tilde{Q}_t(B)}$$

and Stirling's formula prove (4.5), since $N = n - \ell$. $\square$

Remark 4.3 (What is, and is not, independent). The proportions D , the tail Z , and the slack E are independent in the approximating model. The bulk *coordinates* AD_j are not independent of the tail or slack, because their common radius is $\mu - R - E$. This radial constraint is essential. Replacing the entire configuration by the independent tilted product gives distance tending to one, as proved below.

On the exceptional event $\mu - R - E \leq 0$, the model's declared slack E is not equal to $\mu - \sum_j Y_j^*$. This event has probability $O_q(\rho_n)$ by (4.6). The positive-part convention merely defines the model everywhere. One may instead reject that event, changing the approximation by at most another $O_q(\rho_n)$.

Corollary 4.4 (Endpoint slack and a uniform distribution-ratio law). *We have*

$$d_{\text{TV}}(\mathcal{L}_{P_x}(H), \text{Exp}(1)) = O_q(\rho_n), \quad F(x) = \frac{L(t)e^{tx}}{\sqrt{2\pi n}}(1 + O_q(\rho_n)). \quad (4.7)$$

With $F(y) = 0$ for $y \leq 0$,

$$\sup_{u \geq 0} \left| \frac{F(x - u/t)}{F(x)} - e^{-u} \right| = O_q(\rho_n). \quad (4.8)$$

Writing $F^{-1}(v) = \inf\{y : F(y) \geq v\}$ for the generalized inverse, for every fixed $p \in (0, 1)$,

$$F^{-1}(pF(x)) = x + \frac{\log p}{t} + o(t^{-1}). \quad (4.9)$$

All fixed nonnegative integer moments of H converge to those of $\text{Exp}(1)$.

Proof. Projection of (4.4) gives the first statement. The exact normalization identity (2.7) gives the second. The ratio in (4.8) is $P_x(H \geq u)$, so its uniform error is bounded by the total-variation distance. The quantile assertion follows by applying the ratio law at $u = -\log p \pm \eta$ and then letting $\eta \downarrow 0$.

For the moment statement, total variation alone is insufficient. The density of the capped bulk sum under Q_t is bounded by $g_N/P_{\tilde{Q}_t}(B)$; convolution with the tail thus bounds the density f_T of T by $C_q/\sqrt{N}$. Hence the exact slack density satisfies

$$f_H(h) = \frac{e^{-h} f_T(\mu - h)}{D_t} \mathbf{1}_{h \geq 0} \leq C_q e^{-h}.$$

The normalizer estimate and dominated convergence (or uniform integrability combined with weak convergence) now give every stated moment limit. $\square$

This saddle prefactor is deliberately weaker than the all-orders Fabius asymptotics reported in the repository. Its role is to support configuration limits without importing those deeper expansions.

5 Exact marginal likelihoods and a uniform first correction

The full-configuration approximation is strong enough for qualitative limits, but its $O(\log \log n/n)$ error would hide the leading $1/n$ correction for a fixed coordinate. We therefore retain an exact gamma identity before expanding. For $M \geq 1$ and real A ,

$$\int_0^\infty e^{-h} g_M(A - h) dh = g_{M+1}(A). \quad (5.1)$$

For $A > 0$, the exponentials cancel inside the integral and $\int_0^A (A - h)^{M-1} dh = A^M/M$; for $A \leq 0$ both sides are zero.

Define the gamma-at-mode ratio

$$h_M(d) = \frac{g_{M+1}(M + d)}{c_M} = \exp(M \log(1 + d/M) - d) \mathbf{1}_{\{d > -M\}}. \quad (5.2)$$

The value at or to the left of $-M$ is set to zero.

Lemma 5.1 (A globally controlled gamma expansion). *For $M \geq 2$ and every real d ,*

$$0 \leq h_M(d) \leq 1, \quad \left| h_M(d) - 1 + \frac{d^2}{2M} \right| \leq C \frac{|d|^3 + d^4}{M^2}. \quad (5.3)$$

Also, uniformly for z in a fixed compact set,

$$h_M(z\sqrt{M}) \longrightarrow e^{-z^2/2}. \quad (5.4)$$

For $1 \leq k \leq N/2$,

$$\frac{c_{N-k}}{c_N} = 1 + \frac{k}{2N} + O\left(\frac{k^2 + k}{N^2}\right). \quad (5.5)$$

Proof. The inequality $\log(1 + u) \leq u$ proves $h_M \leq 1$. For $|d| \leq M/2$, set $L = M \log(1 + d/M) - d \leq 0$. Taylor's formula gives

$$\left|L + \frac{d^2}{2M}\right| \leq C|d|^3/M^2, \quad |L| \leq Cd^2/M.$$

Since $|e^L - 1 - L| \leq L^2/2$ for $L \leq 0$, (5.3) follows there. For $|d| > M/2$, use $0 \leq h_M \leq 1$ and increase the absolute constant; the term d^4/M^2 controls $1 + d^2/M$. The local limit follows by inserting $d = z\sqrt{M}$ into the logarithm.

For the ratio of constants, use the exact recurrence

$$\frac{c_{j-1}}{c_j} = e(1 - 1/j)^{j-1}.$$

Its logarithm is $1/(2j) + O(j^{-2})$. Summing from $j = N - k + 1$ to N gives $k/(2N) + O((k^2 + k)/N^2)$; exponentiating proves (5.5) uniformly for $k \leq N/2$. $\square$

Fix any $k < N$ bulk coordinates; by symmetry under the uncapped measure we may label them $1, \dots, k$. Put

$$s = y_1 + \dots + y_k, \quad z = s - k, \quad M = N - k.$$

Their likelihood under $\tilde{P}_x$, relative to $\Pi_k = \text{Exp}(1)^{\otimes k}$, is exactly

$$\mathcal{R}_{t,k}(s) = \frac{\mathbb{E}_R g_{M+1}(\mu - s - R)}{\tilde{D}_t} = \frac{c_M \mathbb{E} h_M(-z - W - \varepsilon)}{c_N \mathbb{E} h_N(-W - \varepsilon)}. \quad (5.6)$$

The denominator equality uses (5.1). If the entire tail Z is also observed, its joint likelihood relative to $\Pi_k \otimes Q_t^{>N}$ is instead

$$\mathcal{R}_{t,k}^{\text{tail}}(s, Z) = \frac{g_{M+1}(\mu - s - R)}{\tilde{D}_t} = \frac{c_M h_M(-z - W - \varepsilon)}{c_N \mathbb{E} h_N(-W - \varepsilon)}. \quad (5.7)$$

It is important not to average over R in this latter numerator.

Theorem 5.2 (Universal first bulk correction). *Let $1 \leq k \leq n/3$, and select any k indices in $[1, N]$. Write $\hat{\mathcal{R}}_{t,k}$ for the density of their actual conditioned law relative to Π_k . Then*

$$\hat{\mathcal{R}}_{t,k}(y) = 1 + \frac{k - (s - k)^2}{2n} + \mathcal{E}_{t,k}(y), \quad \int |\mathcal{E}_{t,k}| d\Pi_k \leq C_q \frac{(k + \ell + 1)^2}{n^2}. \quad (5.8)$$

Consequently, if $G_k \sim \text{Gamma}(k, 1)$ and J is the selected set of indices,

$$d_{\text{TV}}(P_x^J, Q_t^J) = \frac{1}{4n} \mathbb{E} |k - (G_k - k)^2| + O_q \left(\frac{(k + \ell + 1)^2}{n^2} \right). \quad (5.9)$$

Proof. Let $\sigma^2 = \mathbb{E} W^2$. Applying Lemma 5.1 to the numerator of (5.6) gives

$$\mathbb{E} h_M(-z - W - \varepsilon) = 1 - \frac{(z + \varepsilon)^2 + \sigma^2}{2M} + u_M(z). \quad (5.10)$$

When s has its reference $\text{Gamma}(k, 1)$ law, z is centered, $\mathbb{E}z^2 = k$, and $\mathbb{E}z^4 = 3k^2 + 6k$. Independence from W , Lemma 3.1, and the bound in (5.3) show that

$$\mathbb{E}_s |u_M(s - k)| \leq C_q \frac{(k + \ell + 1)^2}{M^2}. \quad (5.11)$$

For example, the fourth moment of $z + W + \varepsilon$ is $O_q((k + \ell + 1)^2)$; the third absolute moment is bounded by the fourth-moment estimate and is no larger at the required order.

Similarly,

$$\mathbb{E}h_N(-W - \varepsilon) = 1 - \frac{\sigma^2 + \varepsilon^2}{2N} + O_q\left(\frac{(\ell + 1)^2}{N^2}\right). \quad (5.12)$$

This denominator tends to one. Combine (5.10)–(5.12) with (5.5). To first order the resulting expression is

$$1 + \frac{k}{2N} - \frac{z^2}{2M} - \frac{\sigma^2}{2M} + \frac{\sigma^2}{2N}.$$

The two tail-variance terms differ by $O_q(k(\ell + 1)/N^2)$. Replacing M by N in the z^2 term costs, after integration, $O(k^2/N^2)$. The products of the discarded terms satisfy the same bound as (5.11). Terms involving ε are smaller by (3.5). Thus the integrated error in replacing the likelihood by $1 + (k - z^2)/(2N)$ is $O_q((k + \ell + 1)^2/N^2)$.

Replacing N by n costs at most $Ck\ell/n^2$ in L^1 , since $\mathbb{E}|k - z^2| \leq 2k$. This is absorbed in the stated error. Finally, projection of (3.7) bounds the L^1 difference between the actual and uncapped conditioned densities by $O_q(n^{-11/2})$. This is also absorbed. We have proved (5.8).

The distance between the selected tilted law Q_t^J and Π_k is at most $\sum_{j \in J} e^{-a_j} \leq \delta$. Apply the triangle inequality for total variation and $||a + b| - |a|| \leq |b|$ to obtain (5.9). $\square$

The cancellation of σ^2 in this proof matters. Simply using the global $O(\rho_n)$ coupling would not identify the coefficient in (5.8). The exact gamma-at-mode formula does identify it.

Corollary 5.3 (Sublinear blocks and a one-coordinate constant). *If $k \rightarrow \infty$ and $k = o(n)$, for any k bulk coordinates,*

$$d_{\text{TV}}(P_x^J, Q_t^J) \sim \varphi(1) \frac{k}{n}, \quad \varphi(1) = \frac{e^{-1/2}}{\sqrt{2\pi}} = 0.2419707245 \dots \quad (5.13)$$

For a single bulk coordinate,

$$d_{\text{TV}}(P_x^{\{j\}}, Q_t^{\{j\}}) = \frac{2e^{-2}}{n} + O_q\left(\frac{(\ell + 1)^2}{n^2}\right). \quad (5.14)$$

Proof. The gamma central limit theorem gives $(G_k - k)/\sqrt{k} \Rightarrow Z_0$, where Z_0 is standard normal. Its fourth moments are uniformly bounded, so $\mathbb{E}|1 - (G_k - k)^2/k| \rightarrow \mathbb{E}|1 - Z_0^2| = 4\varphi(1)$. The error in (5.9) is $o(k/n)$: after division it is bounded by a constant times $k/n + \ell/n + (\ell + 1)^2/(nk)$. For $k = 1$, direct integration gives

$$\int_0^\infty |1 - (y - 1)^2| e^{-y} dy = 2 \int_0^2 (2y - y^2) e^{-y} dy = 8e^{-2}.$$

Insert these evaluations into (5.9). $\square$

The growing-block coefficient matches the classical finite exponential-family phenomenon discussed in Section 1. Formula (5.8) specifies how that behavior survives a lacunary infinite tail and a moving small-ball saddle.

Corollary 5.4 (First moment and covariance corrections). *For any fixed number of selected bulk coordinates, uniformly in their indices,*

$$\mathbb{E}_{P_x} Y_i = 1 - \frac{1}{n} + O_q((\ell + 1)^2/n^2), \quad (5.15)$$

$$\text{Var}_{P_x}(Y_i) = 1 - \frac{4}{n} + O_q((\ell + 1)^2/n^2), \quad (5.16)$$

$$\text{Cov}_{P_x}(Y_i, Y_j) = -\frac{1}{n} + O_q((\ell + 1)^2/n^2) \quad (i \neq j). \quad (5.17)$$

Proof. For fixed k , the proof of Theorem 5.2 also holds after multiplication by any fixed polynomial in the observed coordinates and integration of the absolute error. Indeed, the reference exponentials have all fixed moments, and the fourth-degree tail terms are still bounded by $O_q((\ell + 1)^2)$. The cap-error terms are controlled by polynomially weighted exponential tails. Thus the needed moment errors remain $O_q((\ell + 1)^2/n^2)$; this is stronger than an inference from unweighted total variation.

For $k = 1$, the correction is $(2y - y^2)/(2n)$. Using $\mathbb{E}_{\text{Exp}(1)} Y^r = r!$ gives $\mathbb{E}_{P_x} Y = 1 - 1/n + O_q((\ell + 1)^2/n^2)$ and $\mathbb{E}_{P_x} Y^2 = 2 - 6/n + O_q((\ell + 1)^2/n^2)$. For $k = 2$, with $s = Y_1 + Y_2$, the correction is $(2 - (s - 2)^2)/(2n)$. Under the independent reference law,

$$\mathbb{E}(Y_1 Y_2) = 1, \quad \mathbb{E}(Y_1 Y_2 s) = 4, \quad \mathbb{E}(Y_1 Y_2 s^2) = 20.$$

Hence $\mathbb{E}_{P_x}(Y_1 Y_2) = 1 - 3/n + O_q((\ell + 1)^2/n^2)$. Subtracting the appropriate products of means proves the variance and covariance formulas. $\square$

These fixed-dimensional expansions should not be summed over an uncontrolled number of indices to infer a process theorem. That theorem is proved independently in Section 7.

6 The exact threshold for arbitrary coordinate sets

Define a continuous function on $[0, 1]$ by $\mathcal{V}(0) = 0$, $\mathcal{V}(1) = 1$, and

$$\mathcal{V}(\alpha) = 2 \left[\Phi \left(\sqrt{\frac{-\log(1 - \alpha)}{\alpha}} \right) - \Phi \left(\sqrt{\frac{-(1 - \alpha) \log(1 - \alpha)}{\alpha}} \right) \right], \quad 0 < \alpha < 1. \quad (6.1)$$

Also put

$$I(\alpha) = \frac{-\log(1 - \alpha) - \alpha}{2}, \quad 0 \leq \alpha < 1, \quad I(1) = +\infty. \quad (6.2)$$

Theorem 6.1 (Sharp observation-mask criterion). *Let $I = I_t \subseteq \mathbb{N}$ be any deterministic, possibly infinite, coordinate set, and suppose*

$$\alpha_t = \frac{|I_t \cap [1, n]|}{n} \longrightarrow \alpha \in [0, 1].$$

Then

$$d_{\text{TV}}(P_x^{I_t}, Q_t^{I_t}) \longrightarrow \mathcal{V}(\alpha), \quad D_{\text{KL}}(P_x^{I_t} \| Q_t^{I_t}) \longrightarrow I(\alpha). \quad (6.3)$$

In particular, without requiring α_t to have a limit,

$$d_{\text{TV}}(P_x^{I_t}, Q_t^{I_t}) \longrightarrow 0 \iff \frac{\sum_{j \in I_t} \text{Var}_{Q_t}(Y_j)}{\text{Var}_{Q_t}(T)} \longrightarrow 0. \quad (6.4)$$

For initial segments, the criterion is exactly $k = o(n)$; if $k/n \rightarrow 1$ or $\liminf k/n > 1$, their total-variation distance tends to one.

6.1 Proof for a proportional bulk block

First let $J \subseteq [1, N]$ have size k , with $k/n \rightarrow \alpha \in (0, 1)$. Under the uncapped reference law, write $Z_k = (s - k)/\sqrt{k}$. Then $Z_k \Rightarrow Z_0 \sim \mathcal{N}(0, 1)$ and $W/\sqrt{n} \rightarrow 0$ in probability. By (5.4), (5.6), and Stirling's formula,

$$\mathcal{R}_{t,k}(k + \sqrt{k}z) \rightarrow G_\alpha(z) := (1 - \alpha)^{-1/2} \exp\left(-\frac{\alpha z^2}{2(1 - \alpha)}\right) \quad (6.5)$$

uniformly for z in compact sets. To justify the expectation over W , first restrict $|W| \leq \eta\sqrt{n}$, apply the local expansion uniformly on compact z -sets, and then use Chebyshev's inequality and $h_M \leq 1$; finally let $\eta \downarrow 0$.

There is also a uniform upper bound on these likelihoods. Indeed, $h_M \leq 1$, $c_M/c_N \rightarrow (1 - \alpha)^{-1/2}$, and the denominator $\mathbb{E}h_N(-W - \varepsilon)$ tends to one. Tightness of Z_k therefore upgrades (6.5) to

$$\frac{1}{2}\mathbb{E}|\mathcal{R}_{t,k}(s) - 1| \rightarrow \frac{1}{2}\mathbb{E}|G_\alpha(Z_0) - 1|, \quad (6.6)$$

$$\mathbb{E}[\mathcal{R}_{t,k}(s) \log \mathcal{R}_{t,k}(s)] \rightarrow \mathbb{E}[G_\alpha(Z_0) \log G_\alpha(Z_0)]. \quad (6.7)$$

For entropy, use continuity and boundedness of $u \log u$ on a fixed compact interval $[0, C]$, taking $0 \log 0 = 0$.

The same proof works with the entire tail adjoined to J , using (5.7). Now the argument of h_M contains the observed W , rather than an average over it, but $W/\sqrt{n} \rightarrow 0$ gives exactly the same limiting random likelihood.

For completeness, removing the caps does not invalidate the entropy limit. For an observed bulk set J , condition the complementary untruncated bulk on its caps. The change in the conditional expectation of the bounded weight in (2.7) is $O(\delta)$, uniformly in the observed values (and also in an observed tail). After division by $D_t \asymp n^{-1/2}$, the actual likelihood differs by $O_q(\sqrt{n}\delta)$ from its uncapped counterpart on the observed cap set. The observed reference products differ by at most δ , and both likelihoods are uniformly bounded when $k/n \leq \alpha_0 < 1$. Consequently the total-variation and entropy integrals have the same limits. No assertion that entropy is generally continuous in total variation is needed.

When $\alpha = 0$, the same likelihood formulas give convergence to one in probability: $(s - k)/\sqrt{n} \rightarrow 0$ in probability, even if k stays bounded, and $W/\sqrt{n} \rightarrow 0$. The likelihoods remain bounded. The above integration arguments give zero limits for both distances. The case $k = 0$ means observing only the tail and follows directly from (5.7), with $s = 0$ and $M = N$.

6.2 Evaluation of the limiting integrals

The function G_α is the density ratio of $\mathcal{N}(0, 1 - \alpha)$ to $\mathcal{N}(0, 1)$. The ratio exceeds one exactly on $|z| < c_\alpha$, where

$$c_\alpha^2 = -\frac{1 - \alpha}{\alpha} \log(1 - \alpha).$$

Integrating the difference of the two normal densities on this interval proves (6.1). Similarly,

$$\mathbb{E}[G_\alpha(Z_0) \log G_\alpha(Z_0)] = -\frac{1}{2} \log(1 - \alpha) - \frac{\alpha}{2(1 - \alpha)}(1 - \alpha) = \mathcal{I}(\alpha).$$

This also gives

$$\mathcal{V}(\alpha) \sim \varphi(1)\alpha \quad (\alpha \downarrow 0), \quad 1 - \mathcal{V}(\alpha) \sim \sqrt{\frac{2}{\pi}} \sqrt{(1 - \alpha) \log \frac{1}{1 - \alpha}} \quad (\alpha \uparrow 1). \quad (6.8)$$

The first asymptotic follows by Taylor expansion at the two normal arguments near one. For the second, the smaller argument is $\sqrt{(1 - \alpha) \log(1/(1 - \alpha))}(1 + o(1))$, while the upper normal tail at the larger argument is smaller by a logarithmic factor.

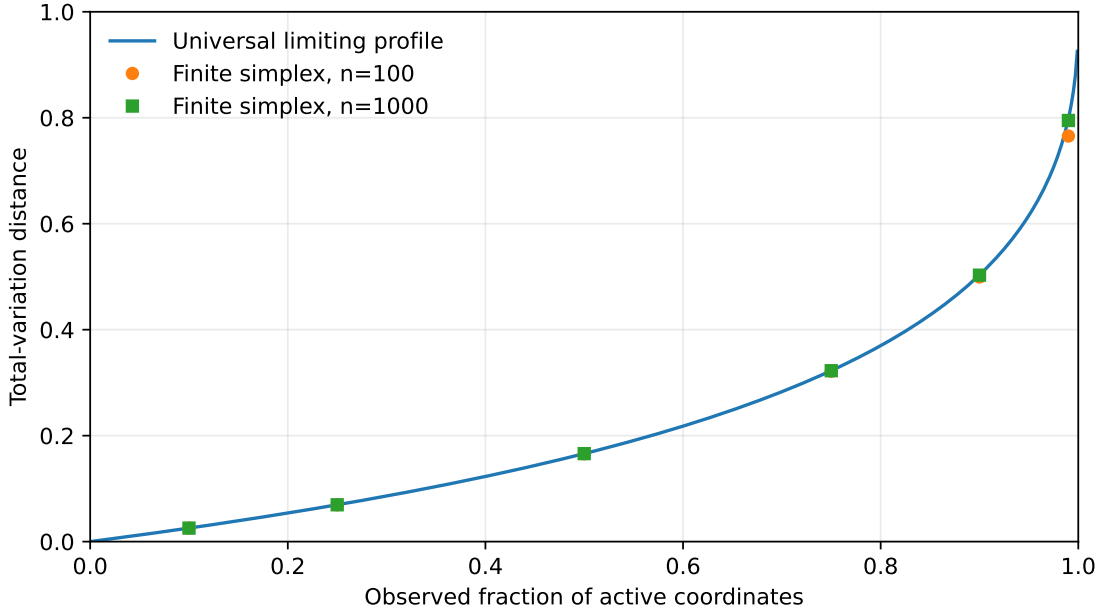


Figure 1: The universal proportional-observation profile (6.1). Markers are deterministic evaluations for the finite uniform-volume simplex proxy described in Section 10; they are not exact Fabius values.

6.3 Proof for an arbitrary, possibly infinite, mask

Let $J = I_t \cap [1, N]$ and $k = |J|$. Then $k/n - \alpha_t \rightarrow 0$ because only $\ell = o(n)$ active indices were removed. Observation of J is a projection of observation of I_t , and observation of I_t is a projection of observation of $J \cup (N, \infty)$. Thus contraction of total variation and the data-processing inequality for relative entropy give a lower and an upper bound using these two extreme observations. For $\alpha < 1$, both extremes have the same limits by the preceding proof. This proves (6.3) for every such α , regardless of how many tail coordinates the mask contains.

For $\alpha = 1$, choose, within J , any $\lfloor \beta n \rfloor$ coordinates with fixed $\beta < 1$. Projection gives a limiting lower bound $\mathcal{V}(\beta)$ for total variation and $\mathcal{I}(\beta)$ for entropy. Letting $\beta \uparrow 1$ proves the asserted limits one and infinity.

Finally, Lemma 2.1 identifies the variance ratio in (6.4) as $\alpha_t + O_q(n^{-1})$. If it tends to zero, the already proved case $\alpha = 0$ applies. Conversely, if α_t does not tend to zero, a subsequence has a positive limiting fraction; its limiting distance is strictly positive. This proves the equivalence and completes Theorem 6.1. $\square$

Corollary 6.2 (Local independence versus global information). *For the full configuration,*

$$d_{\text{TV}}(P_x, Q_t) \longrightarrow 1, \quad D_{\text{KL}}(P_x \| Q_t) = \frac{1}{2} \log(2\pi n) - 1 + o(1). \quad (6.9)$$

Nevertheless every observation mask with negligible tilted variance has vanishing total-variation distance from its independent tilted law.

Proof. The total-variation assertion is the full-mask case of Theorem 6.1. From (2.7),

$$D_{\text{KL}}(P_x \| Q_t) = -\mathbb{E}_{P_x} H - \log D_t.$$

Corollary 4.4 gives $\mathbb{E}_{P_x} H \rightarrow 1$, and (4.5) gives the logarithmic term. The final assertion restates (6.4). $\square$

The theorem shows why a fixed-dimensional Gibbs-conditioning principle alone is incomplete here. It does not identify the largest admissible observation scale, the nonzero proportional-block discrepancy, or the information still present in the full configuration.

7 A Brownian bridge for the active coordinates

Define a continuous process by linear interpolation of the values

$$B_t(j/n) = \frac{1}{\sqrt{n}} \sum_{i=1}^j (Y_i - m(a_i)), \quad j = 0, \dots, n. \quad (7.1)$$

The indexing is by the number of active coordinates, not by their original weights. Exponential tilting is what makes that indexing asymptotically homogeneous.

Theorem 7.1 (Conditioning turns Brownian motion into a bridge). *As random elements of $C([0, 1])$ with the uniform topology,*

$$B_t \Rightarrow B^0 \quad \text{under } P_x, \quad B_t \Rightarrow B \quad \text{under } Q_t, \quad (7.2)$$

where B is standard Brownian motion and $B^0(u) = B(u) - uB(1)$ is a standard Brownian bridge. In particular,

$$\text{Cov}(B^0(u), B^0(v)) = \min(u, v) - uv.$$

The conditioned bridge limit is jointly independent of the limiting exponential slack. In the geometric case it is also jointly independent of every fixed moving boundary window described in Theorem 8.1.

Proof. Use Theorem 4.2, and represent the Dirichlet proportions as $D_i = E_i/G_N$, where the E_i are independent $\text{Exp}(1)$ variables, $G_N = \sum_{i \leq N} E_i$, and these auxiliary variables are independent of the model tail and slack. Let L_N be the linearly interpolated centered partial sum process of the E_i , divided by $\sqrt{N}$.

The elementary exponential invariance principle gives $L_N \Rightarrow B$; a proof is included in Appendix A. The linearly interpolated Dirichlet cumulative process satisfies the exact identity

$$\sqrt{N} \left(\sum_{i \leq Nu}^{\text{interp}} D_i - u \right) = \frac{N}{G_N} (L_N(u) - uL_N(1)). \quad (7.3)$$

Since $G_N/N \rightarrow 1$ in probability, it converges to B^0 .

In the global model, on the event of positive radius,

$$A - N = -W - E - \varepsilon = O_P(\sqrt{\ell + 1}).$$

The exceptional nonpositive-radius event tends to zero. Thus $(A - N)/\sqrt{n} \rightarrow 0$ and $A/N \rightarrow 1$ in probability. Multiplying the Dirichlet process by the radius and centering the first N coordinates by $m(a_i)$ instead of one changes the bulk process by a term tending uniformly to zero: the radial error is at most $|A - N|/\sqrt{n}$, and the total centering discrepancy is at most $\varepsilon/\sqrt{n}$.

For the last ℓ active coordinates, the maximum absolute cumulative contribution after centering is at most $(R + r)/\sqrt{n}$. Under the model this is $O_P((\ell + 1)/\sqrt{n}) = o_P(1)$, by Lemma 3.1. Also $N/n \rightarrow 1$, so the change from the bulk time scale j/N to j/n disappears in the uniform process limit. Total-variation contraction transfers the conclusion from the model to P_x .

Under Q_t , couple the first N coordinates to independent exponentials with failure probability at most δ , and use the same tail and centering bounds. This gives the Brownian-motion assertion.

Finally, the Dirichlet process in (7.3) is a function of the proportions alone and is independent, for each t , of the model slack and tail. Its radial approximation error vanishes in probability. Product weak convergence therefore proves the stated joint independence, once the finite boundary-window convergence is established below. $\square$

This process theorem explains the proportional-block likelihood profile. An observed fraction α of the active sum has asymptotic conditional variance $\alpha(1 - \alpha)$ rather than the tilted variance α . The Gaussian density ratio in (6.5) records exactly that reduction. The full path theorem is stronger than this explanation: it also controls all partial sums simultaneously.

8 Geometric weights: a microscopic phase that does not disappear

Take $w_j = b^{-j}$ with $b > 1$. Then

$$n = \lfloor \log_b t \rfloor, \quad \theta = \{\log_b t\}, \quad a_{n+r} = b^{\theta-r}. \quad (8.1)$$

At indices well before n , the caps diverge and the exponential description applies. Near n , the caps remain of order one and retain the phase θ .

Theorem 8.1 (Phase-dependent boundary law). *Let $J \subset \mathbb{Z}$ be finite. Along any sequence with $\theta \rightarrow \theta_0 \in [0, 1]$, the conditioned boundary window satisfies*

$$\mathcal{L}_{P_x}((Y_{n+r})_{r \in J}, H) \longrightarrow \left(\bigotimes_{r \in J} p_{b^{\theta_0-r}}(y_r) dy_r \right) \otimes \text{Exp}(1) \quad \text{in total variation.} \quad (8.2)$$

For the original uniforms U_{n+r} the limiting density on $[0, 1]$ is

$$u \mapsto \frac{ae^{-au}}{1 - e^{-a}}, \quad a = b^{\theta_0-r}. \quad (8.3)$$

The convergence is joint with the bridge in Theorem 7.1, in the product of the uniform process topology and the finite-dimensional topology; all the displayed limiting components are independent.

Proof. Since $\ell \rightarrow \infty$, every fixed index $n + r$ belongs to the tail beyond $N = n - \ell$ for all sufficiently large n . The global approximation therefore compares the boundary window and slack, with error $O_b(\rho_n)$, to their independent tilted laws. Formula (8.1) gives the exact cap of each coordinate. The truncated-exponential density depends continuously in L^1 on its positive cap, so these tilted laws converge in total variation. For the original uniforms the same argument uses their exact tilted densities $ae^{-au}/(1 - e^{-a})$ on a common interval $[0, 1]$. The process independence was proved by the product construction in Theorem 7.1. $\square$

The endpoint $\theta_0 = 1$ denotes a limit from below; the fractional part itself lies in $[0, 1)$. At the phase reset, the definition of n shifts by one. Thus the laws at phases zero and one are related by a relabeling, not by equality at a fixed offset r .

There is no phase-independent limit for U_n as $x \downarrow 0$ without subsequence selection. For a uniform variable tilted by e^{-au} , its mean has derivative $-\text{Var}(U) < 0$ with respect to a . Hence the distributions in (8.3) at, for example, $a = 1$ and $a = b^{1/2}$ differ. Both phases are realized by taking $t = b^{m+\theta_0}$ and choosing x from the saddle equation. This is an order-one microscopic effect, despite the phase-free Gaussian and exponential macroscopic limits.

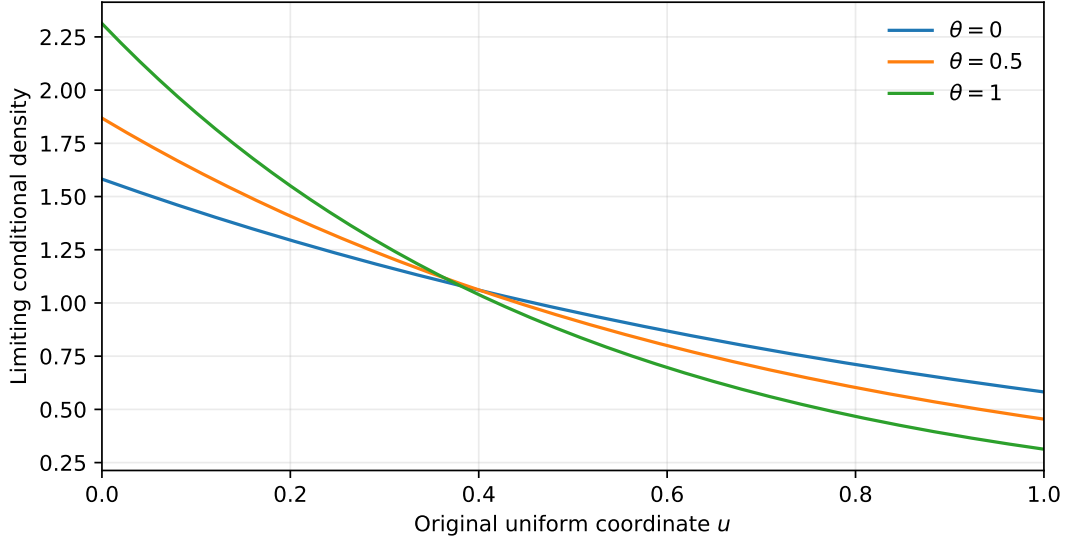


Figure 2: Dyadic boundary densities for U_n , with limiting phase $\theta = 0, 1/2, 1$. The phase-one curve is a limit from below, before the active-index reset. The formula is (8.3), with $a = 2^\theta$.

8.1 An exact phase decomposition of the effective mean

It is useful to distinguish a periodic correction from an index-dependent one. Define, for real θ ,

$$\kappa_b(\theta) = \sum_{r \leq 0} (m(b^{\theta-r}) - 1) + \sum_{r \geq 1} m(b^{\theta-r}). \quad (8.4)$$

Both sums converge locally uniformly, by exponential decay on the left and geometric decay on the right. From the exact index range $r \geq 1 - n$,

$$\mu = n + \kappa_b(\theta) + O_b(te^{-t}). \quad (8.5)$$

Shifting the summation index gives

$$\kappa_b(\theta + 1) = \kappa_b(\theta) + 1.$$

Consequently κ_b itself is *not* one-periodic. The continuous one-periodic correction is

$$\eta_b(\theta) = \kappa_b(\theta) - \theta, \quad \mu = \log_b t + \eta_b(\log_b t) + O_b(te^{-t}).$$

Here η_b is extended periodically. An analogous formula holds for V with m replaced by v and error $O_b(t^2 e^{-t})$. We do not need, and do not assert here, a nonvanishing Fourier coefficient for these particular phase functions. The nonconstant boundary densities already prove the microscopic phase effect directly.

8.2 Translation to the small endpoint parameter

From $tx = n + O_b(1)$ and $\log_b t = n + \theta$,

$$n = \log_b(1/x) + \log_b \log_b(1/x) + O_b(1), \quad t \sim \frac{\log(1/x)}{x \log b}. \quad (8.6)$$

To see this, write $\log_b t = \log_b(1/x) + \log_b(n + O_b(1))$. It first gives $n/\log_b(1/x) \rightarrow 1$, and reinsertion gives the displayed second term. Thus the slack scale is approximately $x \log b / \log(1/x)$, much smaller than x itself. The number of active coordinates is approximately logarithmic in $1/x$.

For $b = 2$, (1.1) is the Fabius distribution in its increasing $[0, 1]$ normalization. One can check the scale without any special-function identity: S_2 has the distribution of $(U + S'_2)/2$, with U independent of a copy S'_2 . Thus

$$F_2(x) = \int_0^1 F_2(2x - u) du,$$

where $F_2 = 0$ below zero and $F_2 = 1$ above one. Differentiation gives $F'_2(x) = 2F_2(2x)$ for $0 < x < 1/2$, and $F_2(1) = 1$.

9 Extremes, empirical laws, and examples beyond the dyadic case

Theorem 9.1 (A Gumbel maximum and an exponential empirical law). *Under P_x ,*

$$\max_{j \geq 1} Y_j - \log n \Rightarrow G, \quad P(G \leq z) = \exp(-e^{-z}). \quad (9.1)$$

For every bounded Lipschitz function $f : [0, \infty) \rightarrow \mathbb{R}$,

$$\frac{1}{n} \sum_{j=1}^n f(Y_j) \longrightarrow \int_0^\infty f(y) e^{-y} dy \quad \text{in } P_x\text{-probability.} \quad (9.2)$$

The maximum limit is jointly independent of the limiting slack and, in the geometric case, a fixed moving boundary window.

Proof. Work first in the global model. With $D_i = E_i/G_N$,

$$\max_{j \leq N} Y_j^* = (A/G_N) \max_{i \leq N} E_i.$$

We have $A/G_N = 1 + O_P(N^{-1/2}) + O_P(\sqrt{\ell+1}/N)$, and $\max_{i \leq N} E_i = \log N + O_P(1)$. Therefore replacing the factor A/G_N by one changes the centered maximum by $o_P(1)$. The elementary identity

$$P\left(\max_{i \leq N} E_i - \log N \leq z\right) = (1 - e^{-z}/N)^N$$

for sufficiently large N proves the Gumbel limit for the bulk.

For the tail, put $h_n = \log n + z$, which eventually exceeds one. Coordinates $j > n$ have caps below one and cannot exceed h_n . There are only ℓ remaining active tail coordinates, and each has tilted exceedance probability at most $2e^{-h_n}$ whenever its cap is at least h_n ; otherwise the probability is zero. Hence

$$Q_t^{>N}(\max_{j > N} Y_j > \log n + z) \leq 2\ell e^{-z}/n \longrightarrow 0.$$

Together with $\log N - \log n \rightarrow 0$ and the global total-variation bound, this proves (9.1).

For (9.2), the first N model coordinates are cE_i with $c = A/G_N \rightarrow 1$ in probability. Their average difference under a Lipschitz function is bounded by

$$\frac{\text{Lip}(f)}{N} |c - 1| \sum_{i \leq N} E_i = o_P(1).$$

The ordinary law of large numbers for $f(E_i)$ applies. The remaining $\ell = o(n)$ terms have a bounded contribution because f is bounded.

For joint independence, use $N \max_i D_i - \log N$, a function of the Dirichlet proportions alone. It has the same Gumbel limit and differs from the radius-rescaled bulk maximum by $o_P(1)$. The proportions are independent of the model slack and tail for every t . Product convergence and Theorem 4.2 finish the proof. $\square$

We have not used or asserted independence between the limiting bridge and the limiting maximum. That stronger joint assertion is a natural extension; it is not necessary for any theorem in this manuscript.

9.1 Three concrete consequences

Observing half the active coordinates. Whether one observes an initial half-block, every other active coordinate, or an arbitrary half of them together with all the inactive tail, the limiting distance from the corresponding independent tilted law is

$$\mathcal{V}(1/2) = 0.16606407498 \dots, \quad \mathcal{I}(1/2) = \frac{1}{2}(\log 2 - 1/2) = 0.09657359028 \dots$$

The vanishing $-1/n$ pair covariance does not imply joint independence of a linear-size block.

Observing infinitely many harmless coordinates. Take $I_t = \{j > n\}$. This is an infinite set, but its total tilted variance is $O_q(1)$, so (6.4) gives $d_{TV}(P_x^{I_t}, Q_t^{I_t}) \rightarrow 0$. The relevant quantity is variance fraction, not cardinality of the observed set.

Supergeometric weights. For $w_j = 2^{-j^2}$, the ratio condition holds, for example, with $q = 1/2$. Now $n = \lfloor \sqrt{\log_2 t} \rfloor$, and $tx = n + O(1)$ gives

$$n \sim \sqrt{\log_2(1/x)}, \quad t \sim \frac{\sqrt{\log_2(1/x)}}{x}.$$

All the global, mask, bridge, and extreme-value theorems still apply, although the number of observable almost-independent coordinates is much smaller than in the Fabius case. A fixed geometric phase is not needed for those theorems. The boundary-window formula, in contrast, specifically uses geometric weights and is not silently extended to this example.

10 Reproducible numerical checks

The package contains `verification.py`, the generated JSON and CSV data, and both figure sources as PDF and PNG images. The computations use floating point; they are not interval-certified inequalities and are not substitutes for the proofs. Two distinct experiments are kept separate.

10.1 An exactly specified finite simplex proxy

Let $(X_1, \dots, X_N)$ be uniform on the volume simplex

$$\{x_i \geq 0 : x_1 + \dots + x_N \leq N\}.$$

For $1 \leq k < N$, its first k coordinates have likelihood relative to $\text{Exp}(1)^{\otimes k}$

$$r_{N,k}(s) = \frac{N!}{(N-k)!N^k} e^s (1-s/N)^{N-k} \mathbf{1}_{\{0 \leq s < N\}}. \quad (10.1)$$

This is a finite proxy, *not* the exact finite truncation of the Fabius conditioning problem. It has the same universal first correction and proportional-block limit, and it is directly computable without Monte Carlo.

The sum s/N under this simplex law has distribution $\text{Beta}(k, N - k + 1)$, whereas s under the reference product has $\text{Gamma}(k, 1)$ distribution. The logarithm of (10.1) is concave and has its maximum at $s = k$. Let s_- and s_+ be the two endpoints of the interval where the likelihood is above one; $s_- = 0$ for $k = 1$. Then the total-variation distance is the beta probability of $[s_-/N, s_+/N]$ minus the gamma probability of $[s_-, s_+]$. The script locates the endpoints by a bracketing solver, using the stable constant

$$\log \frac{N!}{(N-k)!N^k} = \sum_{j=0}^{k-1} \log(1 - j/N).$$

Thus the calculation evaluates an exact distributional formula numerically, rather than estimating a high-dimensional density by sampling.

Table 1: Finite simplex distances and the universal limiting profile.

$\alpha = k/N$	$N = 100$	$N = 1000$	$\mathcal{V}(\alpha)$
0.10	0.025593	0.025498	0.025488
0.25	0.069509	0.069492	0.069491
0.50	0.165814	0.166039	0.166064
0.75	0.321531	0.322560	0.322675
0.90	0.499380	0.502893	0.503283
0.99	0.765440	0.794850	0.798216

For one coordinate, the values of Nd_{TV} at $N = 100, 1000, 10000$ are respectively 0.27157582, 0.27076082, and 0.27067959; their limit is $2e^{-2} = 0.27067057 \dots$. This checks the factor of two in our half- L^1 convention and the fixed-coordinate coefficient independently of a simulation of the infinite series.

10.2 Direct tilted importance sampling for the dyadic series

For $n \in \{32, 64, 128, 256\}$, the script takes $t = 2^{n+0.37}$ and computes the corresponding $x = \mu/t$. It samples the independent tilted coordinates through index $n + 50$ by the inverse distribution formula

$$Y_j = -\log(1 - V_j(1 - e^{-a_j})), \quad V_j \sim \text{Unif}[0, 1]. \quad (10.2)$$

The omitted scaled tail has deterministic upper bound $2^{0.37-50} < 1.2 \cdot 10^{-15}$. With $H = \mu - \sum_j Y_j$ on the sampled truncation, the nonnegative importance weight is $e^{-H} \mathbf{1}_{H \geq 0}$. Weighted ratios estimate the conditioned expectations. There are 240,000 draws at each n , with seeds 20260928 + n .

The standard errors in Table 2 are ordinary importance-sampling standard-error estimates. They describe sampling variability, not finite- n asymptotic error or a rigorous confidence guarantee. In particular the last row gives limiting targets, not the exact expectations at the sampled values of n .

The JSON output additionally reports the boundary-coordinate mean, the slack survival probability at one, the centered-maximum distribution at zero, and effective sample sizes. Finite-size effects are visible: the Gumbel probability at zero is about 0.4012 at $n = 32$, compared with its limit $e^{-1} \approx 0.3679$. This is not hidden by labeling the limiting law an exact finite-size prediction. The deterministic simplex tests pass the script's stated numerical tolerances. The script also checks the polynomial normalization and moment coefficients using exact rational and factorial arithmetic. No formal proof assistant was run.

Table 2: Dyadic importance-sampling checks: estimate $\pm$ one estimated Monte Carlo standard error. The midpoint process is defined in (7.1).

n	$D_t \sqrt{2\pi n}$	$\mathbb{E}_{P_x} H$	$\mathbb{E}_{P_x} B_t(1/2)^2$
32	1.0229 ± 0.0051	0.9685 ± 0.0037	0.2346 ± 0.0017
64	1.0083 ± 0.0062	0.9860 ± 0.0045	0.2398 ± 0.0021
128	0.9953 ± 0.0074	0.9972 ± 0.0054	0.2473 ± 0.0026
256	1.0066 ± 0.0090	0.9890 ± 0.0064	0.2500 ± 0.0032
Limit	1	1	1/4

11 Research questions suggested by the results

The following problems are proposed as continuations of this manuscript. They are not claimed to be previously published open problems unless a separate source establishes that status.

Q1. Is the logarithmic-logarithmic loss necessary? The full-configuration error is $O_q((1 + \log \log n)/n)$ because the removable-cap cutoff leaves $\ell \asymp \log \log n$ coordinates with order-one moments. Can a more accurate radial correction give $O_q(1/n)$ without losing an explicit sampling description? A matching lower bound for the present model would also be useful. One should distinguish optimal approximation of the entire configuration from optimal approximation of a fixed marginal.

Q2. All-order conditional expansions and the first phase coefficient. The exact likelihood (5.6) permits higher-order gamma expansions. The first term is universal after expressing it in n and the scaled coordinates; tail variance cancels. Determine the full second-order term, track the cutoff-independent combination of tail cumulants, and identify where the geometric phase first enters. The repository's periodic saddle coefficient infrastructure is a natural source of comparison, but phase substitution must be justified rather than performed formally.

Q3. Power-law densities at the left endpoint. Replace the uniforms by independent variables with densities $u^{\gamma-1} g_j(u)$ near zero, with $g_j(0) > 0$ and suitable uniform regularity. The tilted bulk should approach gamma variables of shape γ , with Dirichlet($\gamma, \dots, \gamma$) proportions. Establish a full-product coupling, not just fixed-dimensional weak convergence. For varying shapes, determine whether the sharp criterion is the observed fraction of total gamma shape.

Q4. Beyond uniform lacunarity. Weights with $w_{j+1}/w_j \rightarrow 1$ may have a macroscopic transition region rather than $O(1)$ boundary variance. Identify verifiable hypotheses under which a variance-fraction threshold and Gaussian likelihood profile still hold. Polynomial and stretched-exponential weights should be treated separately. A proof must exclude domination by a few coordinates; an analogy with the current active-count argument does not do so automatically.

Q5. Several constraints. Condition on several weighted sums simultaneously. Is there an explicit analogue of the random simplex model, with a finite-rank Gaussian projection in place of the Brownian bridge? Determine which covariance matrix replaces $1 - \alpha$ in the likelihood profile, and how independent exponential slacks are modified when the conditioning region has intersecting faces.

Q6. The almost-full observation regime. Theorem 6.1 gives distance tending to one when the unobserved active fraction vanishes, but it does not give a sharp rate if $n - k$ grows slowly or stays bounded. Analyze the competition between the number of unobserved bulk coordinates, the $\log \log n$ cutoff tail, and the order-one slack. The endpoint expansion in (6.8) alone does not justify an interchange of the proportional and almost-full limits.

Q7. Point-process extremes and joint process limits. Prove convergence of the full extremal point process, with explicit rates, and establish its joint independence from the Brownian bridge where valid. The Dirichlet representation suggests a Poisson limit inherited from exponential order statistics, but the common normalization and the infinite tail must be controlled jointly. The present manuscript proves only the maximum limit and its independence from slack and boundary windows.

Q8. Certified sampling and complexity. Turn the global approximation into a finite algorithm with an explicit user-selected total-variation error. A geometric tail can be truncated with a deterministic norm bound, but truncating coordinates is not automatically small in total variation on the full product space. A correct algorithm must specify its output representation, the observables it approximates, and the metric in which the truncation is certified. Determine the runtime needed for endpoint conditional quantiles or bounded observables.

Q9. Exact shells versus inequalities. Replace $S \leq x$ by a narrow interval $x - h_t \leq S \leq x$, or by a regular conditional law at $S = x$. Identify the transition as th_t tends to zero, a finite constant, or infinity. The exponential slack should be truncated or collapse to zero in the appropriate regimes; the first fixed-coordinate correction should change correspondingly. State the version of conditioning on a null event explicitly.

Q10. Random weights and phase mixing. For random lacunary weights, establish quenched and annealed analogues. Does averaging over a random multiplicative scale erase the geometric boundary phase, and what mixture replaces (8.3)? Such a mixture is not the same as evaluating the phase-dependent density at an average phase. Uniform constants in the deterministic coupling are a useful starting point, but integrability of the random lacunarity parameter needs its own analysis.

Q11. Formal verification of the conditioning layer. A practical first formalization target is the gamma disintegration and the finite-dimensional L^1 expansion, not the full functional limit theorem. One can then connect an infinite-product tilt, countable-product total variation, and cap-removal lemmas to the existing Fabius probability model. The correctness boundary should explicitly distinguish formal inequalities, classical analytic facts imported from a library, and floating-point experiments. No Lean source purporting to certify these new results is included here.

12 A proof-status ledger and conclusions

The central conclusion is a distinction between three scales. A negligible variance fraction is asymptotically unaffected by the global constraint after exponential tilting. A positive fraction retains a universal, exactly quantified dependence. The full configuration retains a logarithmically growing relative-entropy cost and becomes maximally distinguishable from the independent tilted product.

The simplex approximation explains all three scales without discarding the infinite tail. For geometric weights it simultaneously separates universal bulk behavior from a phase-dependent boundary layer. This is

Table 3: Status of the principal components.

Component	Status in this package
Fabius random-series representation	Classical background, cited and normalization checked.
Finite exponential-family variation phenomenon	Classical provenance acknowledged; the needed gamma calculations are rederived.
Full lacunary configuration coupling	Proved in Theorem 4.2, under the explicit ratio assumption.
First marginal correction and mask criterion	Proved in Theorems 5.2 and 6.1; no formal certification claimed.
Bridge, slack, phase, and maximum	Proved with the stated modes of convergence and independence scopes.
Numerical tables	Reproducible floating-point checks, not proofs or rigorous error certificates.
Repository asymptotic audit	Inspected source report; its Lean build and axiom audit were not rerun.
Worldwide novelty / published open problem	No exhaustive priority determination and no claim to resolve a named published conjecture.

the proposed addition to the repository’s Fabius research program: a configuration-level conditioning theory complementing scalar endpoint asymptotics.

A An elementary exponential invariance principle

Let $E_1, E_2, \dots$ be independent $\text{Exp}(1)$ variables, $X_i = E_i - 1$, and let L_N linearly interpolate $N^{-1/2} \sum_{i \leq j} X_i$ at times j/N . We justify the fact $L_N \Rightarrow B$ in $C([0, 1])$ used in Theorem 7.1.

For finite-dimensional distributions, disjoint partial-sum increments are independent. The characteristic function of X_i satisfies

$$\mathbb{E}e^{iuX_i} = 1 - u^2/2 + O(|u|^3) \quad (u \rightarrow 0).$$

For a block of length m_N with $m_N/N \rightarrow a$, its scaled characteristic function tends to $e^{-au^2/2}$. Applying this to finitely many disjoint blocks proves convergence to independent normal increments. Interpolation at finitely many endpoints changes the random variables by $o_P(1)$, since a single $X_i/\sqrt{N}$ tends to zero in probability.

For tightness, independence and the exponential moments give

$$\mathbb{E} \left(\sum_i c_i X_i \right)^4 = 3 \left(\sum_i c_i^2 \right)^2 + 6 \sum_i c_i^4.$$

The increment $L_N(t) - L_N(s)$ has coefficients $c_i/\sqrt{N}$ with $0 \leq c_i \leq 1$ and $\sum_i c_i = N(t-s)$. If $t-s \geq 1/N$, then $\sum_i c_i^2 \leq N(t-s)$ and $\sum_i c_i^4 \leq N(t-s)$; thus its fourth moment is at most $C(t-s)^2$. If $t-s < 1/N$, the sum of the coefficients is at most one, so both terms in the displayed fourth-moment formula are bounded by a constant times $N^4(t-s)^4$. After division by N^2 this is again at most $C(t-s)^2$. We have proved uniformly in N that

$$\mathbb{E}|L_N(t) - L_N(s)|^4 \leq C|t-s|^2.$$

The standard fourth-moment tightness criterion for continuous processes therefore applies. To recall its mechanism, apply Markov’s inequality to dyadic increments, sum their probabilities over each dyadic level, and choose a Hölder exponent below $1/4$; the resulting summable geometric bound controls the modulus of continuity. Finite-dimensional convergence then identifies the unique limit as standard Brownian motion.

B Reproduction and source record

The archive is self-contained for PDF compilation: the LaTeX source uses an embedded bibliography and the included figure PDFs. No online source fetch is performed by the build. A standard TeX Live installation with `newtx`, `microtype`, and the usual AMS packages suffices.

```
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
python verification.py --draws 240000
```

The numerical script requires Python 3.11 or later, NumPy, SciPy, and Matplotlib. The delivered run used NumPy 2.3.5, SciPy 1.17.0, and Matplotlib 3.10.8. The output includes all figures and `data/verification_results.json`. The optional `-skip-monte-carlo` flag reruns only the deterministic simplex checks. The Makefile provides equivalent build targets.

The repository snapshot was obtained from its recursive tree and the asymptotic audit was read at the pinned commit. The bibliography points to that pinned audit, rather than an unspecified future branch state. Narrow repository searches for Gibbs conditioning and Brownian-bridge terminology were used to choose a research direction, not to certify absence of overlap. The cited 2026 version of Cator–Ferrari was checked on arXiv; the Diaconis–Freedman comparison is supported by the publisher’s abstract and bibliographic record. The proofs here do not rely on an inaccessible theorem statement from that paper.

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